

Mach-Zehnder Interferometer Design Simulation and Manufacturing

Hai Nguyen

July 3, 2026

1 Abstract

In this paper, we describe the design process of a Mach-Zehnder interferometer, from simulation to manufacturing. We explore the effects of varying L, from theoretical models to layout designs, and finally to the fabricated results.

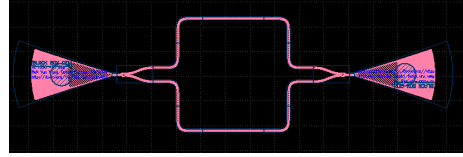


Figure 1: This is an actual layout used for the design and fabrication of a balanced MZI

How does the MZI work? Consider the following figure 2.

2 Introduction

Photonics has become a leading field of study in quantum computing. This introductory course uses one of the fundamental devices, the Mach-Zehnder interferometer, to introduce the principles of photonics. The integration of photonics and semiconductors is a rapidly growing area. This experiment helps us understand the process of designing a Mach-Zehnder interferometer and comparing theoretical data with real-world variations on silicon.

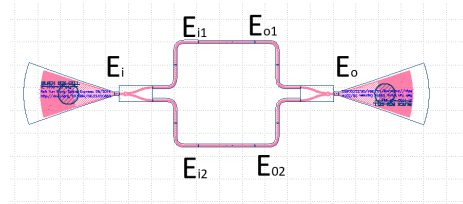


Figure 2: MZI diagram

We see the input energy is E_i and the output is E_o . In a balanced MZI each branch will have the following equations for the inputs of each branch.

$$E_{i1} = E_{i2} = \frac{1}{\sqrt{2}} E_i$$

The output of each branch have the equation: -

$$E_{o1} = E_{i1} * e^{(-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1)}$$

$$E_{o2} = E_{i2} * e^{(-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2)}$$

3 Theory

The Mach-Zehnder Interferometer (MZI) is a 2-port device. It is formed by a beam-splitter and a beam combiner with 2 waveguides in between, see Figure 1.

Where $\beta_{1,2}$ is the propagation constants of the upper and lower branches.

$$\beta_{1,2} = \left(\frac{2\pi}{\lambda}\right)neff_{1,2}$$

The beam-combiner's equation is:

$$E_o = \frac{1}{\sqrt{2}} * (E_{o1} + E_{o2})$$

For an unbalanced MZI, we can substitute equations 1-4 in equation E_o to get:

$$E_o = \frac{1}{2} E_i (e^{(-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1)} + e^{(-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2)})$$

The light intensity is proportional to the squared power of the field amplitude. This gives us:

$$I_o = \frac{1}{4} |I_i| (e^{(-j\beta_1 L_1 - \frac{\alpha_1}{2} L_1)} + e^{(-j\beta_2 L_2 - \frac{\alpha_2}{2} L_2)})^2$$

This equation represents an unbalanced MZI. For our project we can assume there is no propagation loss relative to 1 and 2 and thus the equation can be simplified even further.

$$I_o = \frac{1}{2} I_i (1 + \cos(\frac{\beta}{L}))$$

In our project we will be measuring the period between peaks of the MZI transfer function (FSR) which is calculated with the following equation.

$$FSR(\lambda) = \frac{\lambda^2}{\Delta L * ng(\lambda)}$$

The group index is defined as ng.

$$ng(\lambda) = neff(\lambda) - \lambda \frac{dneff(\lambda)}{d\lambda}$$

4 Modelling and Simulation

In our experiment, we set the $\Delta L = L_2 - L_1$ to be 30, 40, 50, 60, 70, 80, 100, 110, 160, and 200. In this project, we will be using a simple y-branch for the splitter and the combiner. We will have a balanced MZI as a control and the rest of the design will be imbalanced MZI as seen figure 3.

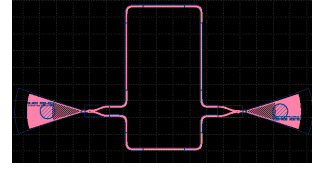


Figure 3: Imbalanced MZI

This layout design was chosen compared to other designs to make it easier to calculate the exact difference between the 2 branches. As seen in Figure 4, only the vertical waveguide (WG) is varying and all other dimensions on the WG are fixed.

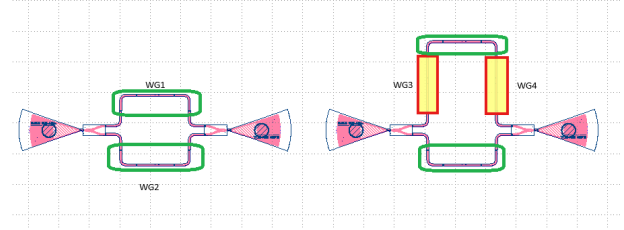


Figure 4: figure on the left shows WG1 and WG2 as fixed dimensions, the figure on the right only have the vertical WG3 and WG4 as variables. WG1 and WG2 are fixed

We will use the Lumerical Interconnect to simulate the theoretical FSR vs and re-run the same simulation on the layout to see if there are any layout dependencies. We will then test the silicon to compare it with the model. Figure 5 is the setup for the theoretical simulation.

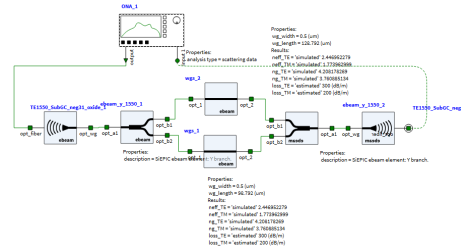


Figure 5: Lumerical CONNECT

For the layout simulation we will use Klayout and

Lumerical Connect to simulate the model from the layout. Below is the table, table 1, of the FSR data of the theoretical model vs the model simulated from the layout

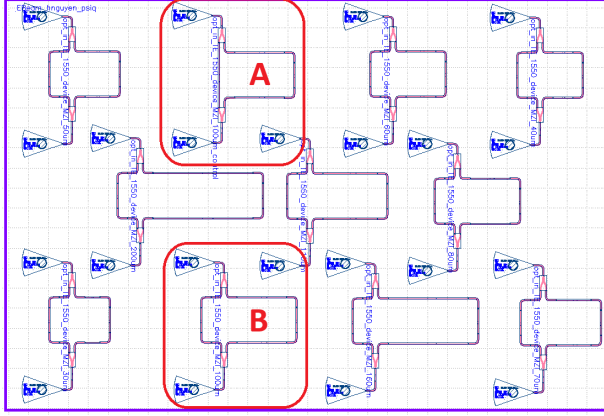


Figure 6: Layout in Klayout used with Lumerical Connect package to simulate layout data.

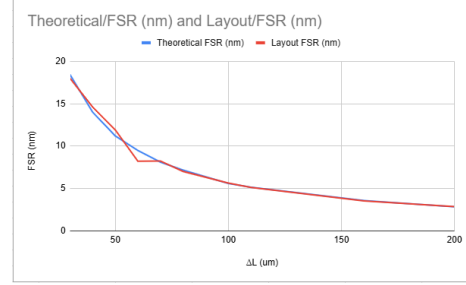


Figure 7: Theoretical vs Layout Model simulations

As you can see for lower ΔL the FSR is off but for larger ΔL greater than 50um, the data between the theoretical and layout tracks well. It will be interesting to see if the silicon data will show the same, considering the numbers of bends in the layout. We also placed a traditional layout to compare and this can be seen in figure 6b. The ΔL is 100um and the simulation shows that there is no change between layout A and B in figure 6. We will now refer figure 6a as the traditional layout and figure 6b as the new layout.

$\Delta L(\mu m)$	Theoretical FSR(nm)	Layout FSR(nm)
30	18.4	18
40	14	14.6
50	11.2	11.9
60	9.48	8.21
70	8.1	8.23
80	7.17	7.01
100	5.6	5.65
110	5.14	5.12
160	3.59	3.53
200	2.83	2.86

Table 1: Theoretical vs Modeling data

$\Delta L(\mu m)$	Theoretical FSR(nm)	Layout FSR(nm)
30	18.4	18
40	14	14.6
50	11.2	11.9
60	9.48	8.21
70	8.1	8.23
80	7.17	7.01
100	5.6	5.65
100T	5.6	5.65
110	5.14	5.12
160	3.59	3.53
200	2.83	2.86

Table 2: Traditional data FSR value

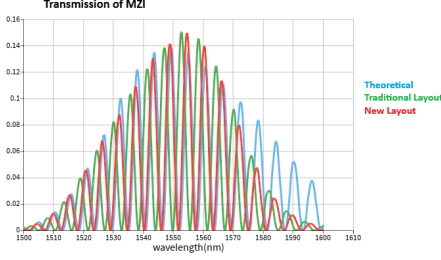


Figure 8: Theoretical vs Traditional vs newlayout $\Delta L=100\mu m$

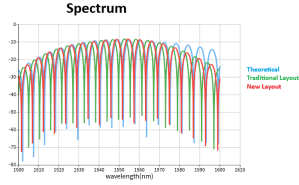


Figure 9: Theoretical vs Traditional vs newlayout $\Delta L=100\mu m$

The differences between the traditional and new layout can be explained. The new layout only has the vertical legs, figure 4, changed and the ΔL can be calculated as integer (μm). Whereas the traditional layout, there are some round errors for the ΔL because we have to take into account the circumference of the corners ($18.448\mu m$). It will be interesting to see the silicon data shows any difference between the traditional and new layout.

5 Analysis

$\Delta L(\mu m)$	Theoretical FSR(nm)	Layout FSR(nm)	Silicon FSR(nm)
30	18.4	18	
40	14	14.6	1.2
50	11.2	11.9	0.69
60	9.48	8.21	9.75 (D2)
70	8.1	8.23	2.28
80	7.17	7.01	1.2 (D3)
100	5.6	5.65	2.69 (D3)
100T	5.6	5.65	4.64
110	5.14	5.12	5.66 (D2)
160	3.59	3.53	5.29 (D3)
200	2.83	2.86	1.08 (D3)

Table 3: FSR Model vs Silicon, D2 = Detector2, D3 = Detector3

6 Conclusion

The data from the experiment was inconclusive. There were too much noise in the measurements to make any definite conclusion.

7 Acknowledgements

I/We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Enxiao Luan performed the measurement